# Exit Ticket · reflect and retrieve

## Exit Ticket

O(n²) means an algorithm takes n² seconds.

## Use key vocabulary from memory

1. Explain Big O in your own words.

2. Explain O(1) in your own words.

3. Explain O(log n) in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 207 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Complexity and algorithm choice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why is a queue suitable for a print server?

### Q2

What condition must the array satisfy?

### Q3

How many times does a for-loop from i=0 to i<6 execute?

### Q4

A loop of length 4 inside another loop of length 4: how many inner operations?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Start with n — n is the number of items. More items usually mean more operations.

## Core idea 2

Trace loop nesting — One full scan performs n steps. A nested n-by-n scan performs $n^2$ steps.

## Core idea 3

Compare growth — When n doubles, linear work doubles; quadratic work roughly quadruples. Big O describes growth, not seconds.

**Key visual · complete concept**

**Complexity and algorithm choice**

| | | |
|---|---|---|
| **n = 8** | **Linear 8** | **Quadratic 64** |
| **n = 16** | **Linear 16** | **Quadratic 256** |
| **n = 32** | **Linear 32** | **Quadratic 1024** |

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Why may linear search be preferable for a single search of a short unsorted array?

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Give the Big O of a single full scan of n elements.

## Question 2

For n=100, give iterations of a linear n loop and nested n×n loop.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

Big O — Asymptotic growth of algorithm work.

O(1) — Constant growth.

O(log n) — Logarithmic growth, e.g. halving search intervals.

O(n) — Linear growth.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Compare operation counts of one loop and two nested loops for n=5.

## Follow the worked method

1. Predict — One loop visits five elements, so does five operations.
2. Explain — The nested loop repeats five operations five times, making 25.

**Source model · final illustrated step**

**Compare operation counts of one loop and two nested loops for**

| | | |
|---|---|---|
| **SINGLE** | **n=5** | **5 operations** |
| **NESTED** | **5×5** | **25 operations** |

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict
Linear search can check all 64 values.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.